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Project Title:	Regulation of Auxin Transport by Phosphorylation and Flavonoids during Gravitropism in Arabidopsis		
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Although observations of the ability of plants to change their growth and developmental patterns in response to gravity date back to Darwin, the mechanisms which control this process are still not understood. One requirement for gravitropic bending is the redistribution of the plant hormone auxin. The potential for understanding how auxin transport is regulated during gravitropic bending has increased dramatically with the isolation of Arabidopsis mutants with defects in genes encoding both auxin transport proteins and regulatory proteins. The goal of this proposal will be to examine two potential mechanisms for regulation of auxin transport during gravitropism. The experimental approach will combine the use of Arabidopsis plants with mutations in genes that alter the regulation of auxin transport and gravity response and inhibitor treatments that mimic these mutant phenotypes; transgenic plants that provide information on changes in gene expression; and computer aided analysis of plant gravity response.
The proposed experiments will examine synthesis of compounds that act to change the activity of auxin transporters and

Task Description:	examine alterations in phosphorylation states that regulate auxin transport during plant gravity response. Preliminary evidence indicates that mutants that synthesize no flavonoids or that have reduced phosphatase activity exhibit significant delays in gravitropic bending. Both in vitro and in vivo evidence indicate that flavonoids are natural regulators of auxin transport. The localization of expression of flavonoid biosynthetic enzymes and flavonoid accumulation will be examined in concert with measurements of gravitropism to determine if the amount or distributions of these compounds affect gravitropic bending. Additionally, the role of phosphorylation in regulation of auxin transport will be examined using Arabidopsis plants with mutations that affect phosphatase activity. Finally, the possibility that phosphorylation and flavonoid synthesis are interconnected regulatory mechanisms will be examined. Phosphorylation states of cellular proteins may affect the transcription of genes encoding flavonoid biosynthetic enzymes or flavonoids may act as kinase inhibitors. Together these experiments should provide a clear sense of how flavonoids and phosphorylation act to regulate auxin transport and why alterations in either flavonoid synthesis or phosphorylation reduce the ability of plants to respond to gravity. The goal of this proposal will be to examine two potential mechanisms for regulation of auxin transport during gravitropism. The proposed experiments will examine synthesis of compounds that act to change the activity of auxin transporters and examine alterations in phosphorylation states that regulate auxin transport during plant gravity response. The proposed and completed experiments provide new insight into the mechanisms by which plants control their growth and response to gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The goal of the research supported by this grant is to increase understanding of a basic biological process, the response of plants to gravity. This study focuses on the role of transport of a class of plant hormones, the auxins, in gravity response and explores how this transport affects plant growth and development. This study will also examine polarity in the distribution of proteins, polarity in cells and tissues, and the polar movement of auxin in order to determine the relationship between these processes and the changes in these processes during gravity response. The long-term goals of this study include understanding how one of the proteins which controls auxin transport functions and how gravity stimuli may affect this protein"s function to allow changes in auxin transport and changes in growth in response to gravity. A clearer understanding of the mechanisms by which plants respond to changes in the vector of gravity may provide important insight into predicting how plants will grow in the absence of gravity and may facilitate the design of experiments to study plant response during space flight.
Task Progress:	The efforts of the past year have focused on both understanding how phosphorylation signaling and flavonoids may control auxin transport to allow it to change during gravity response. We have expanded our studies on mutants with altered phosphorylation signaling to include both the rcn1 mutant, which has a defect in a protein phosphatase regulatory subunit, and the pinoid (pid9) mutant, which has a defect in a kinase that has been suggested to regulate auxin transport and/or auxin signaling. Both mutants have defects in polar auxin transport, with elevated transport in the phosphatase mutant and reduced transport in the kinase mutant, suggesting reciprocal regulation of auxin transport by phosphorylation signaling. Surprisingly, roots of both plants have reductions in gravitropic bending. Our results suggest that elevated auxin transport in the rcn1 mutant overwhelms either the formation or response to gradients in auxin across a root tip needed for gravity response (Rashotte et al. 2001). This is not inconsistent with the reduction in gravity response in the pid9 mutant, as reduced auxin transport may also reduce the speed of formation of an auxin gradient. We are currently investigating other tissues to determine if there are altered gravity responses and auxin transport in the hypocotyl and inflorescences of rcn1. The hypocotyls of rcn1 are phenotypically different from wild-type, when grown in the dark. They also have elevated auxin transport like the roots, but in contrast to the roots, they have accentuated gravitropic bending. These results suggest that phosphorylation signaling may regulate auxin transport in multiple tissues, but that elevated polar transport may have different effects in the hypocotyl than root. This difference may be explained by different mechanisms for lateral redistribution of auxin at the root tip and in the hypocotyl. In the root tip, gravity is sensed by a small number of cells. Redistribution of auxin to the sides may occur in these cells and be very local. In contrast, in hypocotyls, statoliths sediment in cells along the length of the organ and auxin may also redistribute laterally along this entire length. Therefore, elevated transport may more readily overwhelm the gradient at the root tip, than along the hypocotyl. In contrast, more polar transport in the hypocotyl would allow more auxin to redistribute along the length and thereby accentuate the gravity response, as the data indicates. Finally, we are exploring the sensitivity of hypocotyl growth and gravitropism to ethylene in wild-type and rcn1, and are in the final stages of preparing a manuscript on this topic. We have been carefully examining the role of flavonoids as endogenous regulators of auxin transport during gravitropism. The tt4 mutant, which has a mutation in a flavonoid biosynthetic enzyme, makes no flavonoids. Auxin transport is elevated in inflorescences of this mutant, consistent with flavonoids acting as endogenous negative regulators (Brown et al. 2001). In roots of the tt4 mutant, both auxin transport and gravity response are altered. Basipetal IAA transport is elevated in this mutant, consistent with the absence of a negative regulator (Buer and Muday, 2004). The gravity response in tt4 are much slower in initiating gravitropic bending, although later stages of response are wild-type in rate and magnitude. This lag in initiation of bending has been detected by the computer program Multi-ADAPT, which images roots and quantifies bending in real time and with high spatial resolution. As this program only images one root at a time, additional analyses have been performed with multiple plants and the rate of bending has been analyzed by computerized image analysis. Finally, this gravity defect can be chemically complemented by growth of the tt4 mutant on naringenin, a flavonoid intermediate. Since the absence of flavonoids delays gravitropic bending, we have been examining flavonoid synthesis in wild-type plants after gravity stimulation in order to determine if there are changes in response to this stimulus. This experiment is possible using a compound that fluoresces in response to binding of certain flavonoids. We have found that reorientation of plants changes the abundance of flavonoids in the epidermal cells at the tip of the root using confocal microscopy (Buer and Muday, 2004). Currently, we are examining the changes in expression of the gene encoding chalcone synthase, the enzyme that catalyzes the rate limiting step in flavonoid synthesis to determine if gravity induces accumulation of flavonoids due to elevated transcription of the CHS gene. A third set of experiments supported by this grant examines the transport of the endogenous auxin indole-3-butyric acid (IBA). Although indole-3-acetic acid (IAA) is the best studied auxin, it is not clear how similar action and transport of these two auxins are. IBA is abundant in Arabidopsis and has been implicated in gravity response, by isolation of a mutant in the laboratory of Candace Waddell. The rib1 mutant is IBA insensitive, IAA sensitive and impaired in root gravitropism. The transport of IBA in roots, hypocotyls, and inflorescence tissues was quantified and compared to IAA

	movements and the resulting data was recently published (Rashotte et al., 2003). IBA movement is directional and tissue specific and in the roots and hypocotyls is greater than IAA movement. In contrast there is no IBA movement in the inflorescence. Finally, IBA movement does not depend upon the presence of either the AUX1 or EIR1 proteins, two proteins that are required for IAA transport. Additionally, IBA transport is not regulated by IAA efflux inhibitors, suggesting that different proteins mediate IBA movment. We have also examined IBA and IAA transport in the agravitropic rib1 mutant. This mutant has altered IBA transport, but wild-type levels of IAA transport. These results link IBA transport to the response of gravity and this paper is now complete and ready for submission (Poupart et al., in preparation). The final set of experiments that have been supported by this grant are collaborative experiments with Dr. Zhenbiao Yang. His laboratory is examining the effect of brassinolide on lateral root development. We have performed parallel experiments to demonstrate that brassinolide increases auxin transport at concentrations that stimulate lateral root development. These results were recently published (Boa et al., 2004).
Bibliography Type:	Description: (Last Updated: 08/25/2020)
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Presentation	Sun, H.; Basu, S.; Muday, G.K. "Interactions between auxin transport and the actin cytoskeleton." Plant and Fungal Cytoskeleton Gordon Conference. Aug-2004
Presentation	Buer, C.S.; Lovin, M.B.; Muday, G.K. "Environmental modulation of root branching by changing auxin flow from the shoot." 4th International Symposium on Adventitious Root Formation. May-2004